

Experiment/Case Study No : 7

Aim:

Perform DOM traversal and updates; develop a to do list app with add, edit, and delete features.

Software/Tools Required

- Visual Studio Code
- Google Chrome (Web Browser)

Experiment Program Code

1. Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>To-Do List</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="todo-container">
    <h1>To-Do List</h1>
    <div class="input-section">
      <input type="text" id="todo-input" placeholder="Add a new task...">
      <button id="add-btn">Add</button>
    </div>
    <ul id="todo-list"></ul>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

2. style.css

```
body {
  font-family: Arial, sans-serif;
  background-color: #ace768;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
  margin: 0;
}

.todo-container {
  background: #ffffff;
  padding: 2rem;
  border-radius: 8px;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
  width: 100%;
```

```

    max-width: 400px;
}

h1 {
  margin-top: 0;
  color: #333;
  text-align: center;
}

.input-section {
  display: flex;
  gap: 10px;
  margin-bottom: 1.5rem;
}

#todo-input {
  flex: 1;
  padding: 10px;
  border: 1px solid #ccc;
  border-radius: 4px;
  font-size: 16px;
}

#add-btn {
  padding: 10px 16px;
  background-color: #28a745;
  color: white;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  font-size: 16px;
}

#add-btn:hover {
  background-color: #218838;
}

ul {
  list-list-type: none;
  padding: 0;
  margin: 0;
}

li {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background: #f9f9f9;
  padding: 10px;
  border: 1px solid #eee;
  border-radius: 4px;
  margin-bottom: 10px;
}

.task-text {
  flex: 1;
  font-size: 16px;
  color: #333;
}

.task-text[contenteditable="true"] {
  background: #fff;
  border: 1px solid #007bff;
  outline: none;
  padding: 2px 6px;
  border-radius: 4px;
}

```

```

}

.actions {
  display: flex;
  gap: 5px;
}

.edit-btn, .delete-btn {
  padding: 6px 10px;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  font-size: 14px;
}

.edit-btn {
  background-color: #ffc107;
  color: #212529;
}

.edit-btn:hover {
  background-color: #e0a800;
}

.delete-btn {
  background-color: #dc3545;
  color: white;
}

.delete-btn:hover {
  background-color: #bd2130;
}

```

3. Script.js

```

const todoInput = document.getElementById('todo-input');
const addBtn = document.getElementById('add-btn');
const todoList = document.getElementById('todo-list');

addBtn.addEventListener('click', addTask);
todoInput.addEventListener('keypress', function (e) {
  if (e.key === 'Enter') {
    addTask();
  }
});

function addTask() {
  const taskText = todoInput.value.trim();
  if (taskText === "") return;

  const li = document.createElement('li');

  const textSpan = document.createElement('span');
  textSpan.classList.add('task-text');
  textSpan.innerText = taskText;

  const actionsDiv = document.createElement('div');
  actionsDiv.classList.add('actions');

  const editBtn = document.createElement('button');
  editBtn.classList.add('edit-btn');
  editBtn.innerText = 'Edit';

```

```

const deleteBtn = document.createElement('button');
deleteBtn.classList.add('delete-btn');
deleteBtn.innerText = 'Delete';

actionsDiv.appendChild(editBtn);
actionsDiv.appendChild(deleteBtn);
li.appendChild(textSpan);
li.appendChild(actionsDiv);
todoList.appendChild(li);

todoInput.value = "";

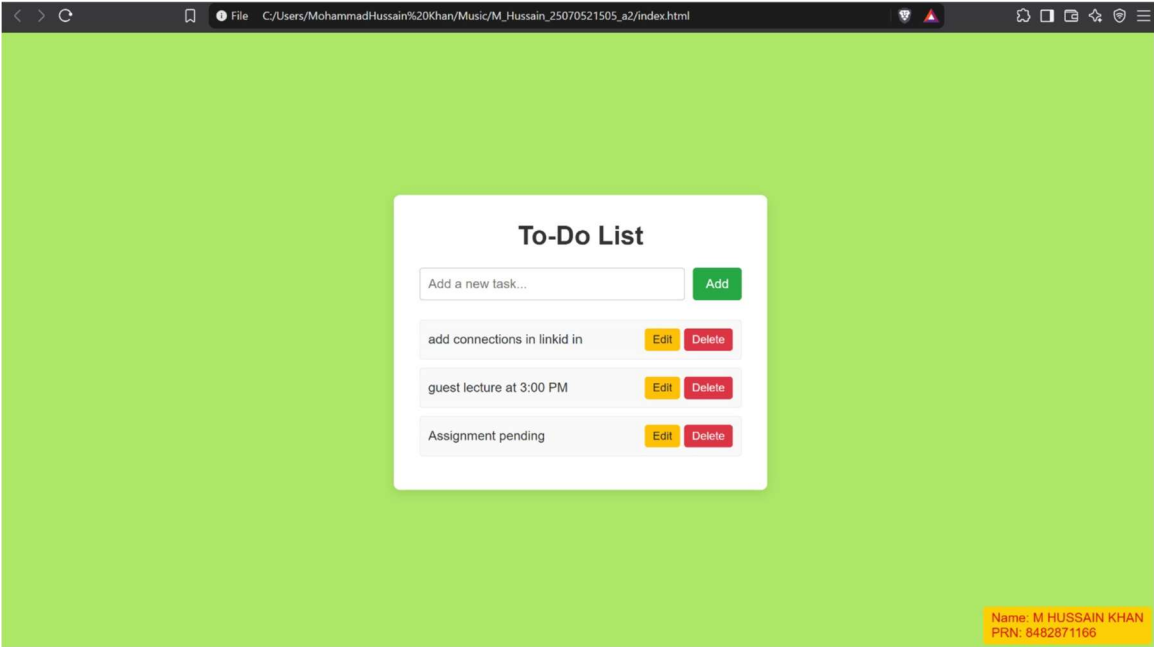
deleteBtn.addEventListener('click', function () {
  const itemToDelete = this.parentElement.parentElement;
  todoList.removeChild(itemToDelete);
});

editBtn.addEventListener('click', function () {
  const itemToEdit = this.parentElement.parentElement;
  const span = itemToEdit.querySelector('.task-text');

  if (this.innerText === 'Edit') {
    span.contentEditable = true;
    span.focus();
    this.innerText = 'Save';
    this.style.backgroundColor = '#007bff';
    this.style.color = 'white';
  } else {
    span.contentEditable = false;
    this.innerText = 'Edit';
    this.style.backgroundColor = '#ffc107';
    this.style.color = '#212529';
  }
});
}

```

Output



Case Study Title

create a registration form.

Case Study Program Code

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Hackathon Registration Form</title>
```

```
  <style>
```

```
    body {
```

```
      font-family: Arial, sans-serif;
```

```
      margin: 40px auto;
```

```
      background-color: #00aeef;
```

```
      max-width: 500px;
```

```
    }
```

```
    .form-container {
```

```
      background: white;
```

```
      padding: 30px;
```

```
      border-radius: 10px;
```

```
      box-shadow: 0 4px 12px rgba(0,0,0,0.1);
```

```
    }
```

```
    h2 {
```

```
      margin-top: 0;
```

```
      color: #333;
```

```
      text-align: center;
```

```
    }
```

```
    .form-row {
```

```
      display: flex;
```

```
    gap: 15px;
}
.form-group {
    margin-bottom: 18px;
    flex: 1;
}
label {
    display: block;
    margin-bottom: 6px;
    font-weight: 600;
    font-size: 0.9em;
    color: #444;
}
input, select, textarea {
    width: 100%;
    padding: 10px;
    border: 1px solid #ccc;
    border-radius: 5px;
    box-sizing: border-box;
    font-size: 14px;
}
input:focus, select:focus, textarea:focus {
    border-color: #007bff;
    outline: none;
    background-color: #fafafa;
}
.error-message {
    color: #dc3545;
    font-size: 0.8em;
    margin-top: 4px;
```

```
        display: none;
    }
    .gender-group {
        display: flex;
        gap: 15px;
        margin-top: 5px;
    }
    .gender-group label {
        font-weight: normal;
        display: flex;
        align-items: center;
        gap: 5px;
        cursor: pointer;
    }
    .gender-group input {
        width: auto;
    }
    .terms-group {
        display: flex;
        align-items: flex-start;
        gap: 8px;
        margin-bottom: 20px;
    }
    .terms-group input {
        width: auto;
        margin-top: 3px;
    }
    .terms-group label {
        font-weight: normal;
    }
```

```
button {
  width: 100%;
  padding: 12px;
  background-color: #007bff;
  color: white;
  border: none;
  border-radius: 5px;
  font-size: 16px;
  font-weight: bold;
  cursor: pointer;
  transition: background 0.2s;
}
button:hover {
  background-color: #b3f811;
}
.success-message {
  color: #28a745;
  font-weight: bold;
  margin-top: 15px;
  text-align: center;
  display: none;
}
</style>
</head>
<body>

<div class="form-container">
  <h2>Hackathon Registration Form</h2>
  <form id="registrationForm" novalidate>
```

```
<div class="form-row">
  <div class="form-group">
    <label for="firstName">First Name *</label>
    <input type="text" id="firstName" placeholder="John">
    <div class="error-message" id="firstNameError">First name is
required.</div>
  </div>
  <div class="form-group">
    <label for="lastName">Last Name *</label>
    <input type="text" id="lastName" placeholder="Doe">
    <div class="error-message" id="lastNameError">Last name is
required.</div>
  </div>
</div>

<div class="form-group">
  <label for="birthday">Birthday *</label>
  <input type="date" id="birthday">
  <div class="error-message" id="birthdayError">Please select a valid
date.</div>
</div>

<div class="form-group">
  <label for="username">Username *</label>
  <input type="text" id="username" placeholder="johndoe123">
  <div class="error-message" id="usernameError">Username must be at least
4 characters long.</div>
</div>

<div class="form-group">
  <label for="email">Email Address *</label>
  <input type="email" id="email" placeholder="john.doe@example.com">
```

```
<div class="error-message" id="emailError">Please enter a valid email address.</div>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="phone">Phone Number *</label>
```

```
<input type="tel" id="phone" placeholder="1234567890">
```

```
<div class="error-message" id="phoneError">Phone number must be exactly 10 digits.</div>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Gender *</label>
```

```
<div class="gender-group">
```

```
<label><input type="radio" name="gender" value="male"> Male</label>
```

```
<label><input type="radio" name="gender" value="female">  
Female</label>
```

```
<label><input type="radio" name="gender" value="other">  
Other</label>
```

```
</div>
```

```
<div class="error-message" id="genderError">Please select your gender.</div>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="country">Country *</label>
```

```
<select id="country">
```

```
<option value="">Choose Country</option>
```

```
<option value="US">United States</option>
```

```
<option value="IN">India</option>
```

```
<option value="UK">UK</option>
```

```
<option value="CA">Canada</option>
```

```
<option value="CA">Japan</option>
```

```
<option value="CA">Australia</option>
<option value="CA">New Zealand</option>
<option value="CA">China</option>
```

```
</select>
```

```
<div class="error-message" id="countryError">Please select a
country.</div>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="bio">Short Bio</label>
```

```
<textarea id="bio" rows="3" placeholder="Tell us about
yourself..."></textarea>
```

```
</div>
```

```
<div class="terms-group">
```

```
<input type="checkbox" id="terms">
```

```
<label for="terms">I agree to the Terms and Conditions *</label>
```

```
</div>
```

```
<div class="error-message" id="termsError" style="margin-top: -15px; margin-
bottom: 15px;">You must accept the terms.</div>
```

```
<button type="submit">Register Now</button>
```

```
<div class="success-message" id="successPrompt">Account created
successfully!</div>
```

```
</form>
```

```
</div>
```

```
<script>
```

```
const form = document.getElementById('registrationForm');
```

```
const firstName = document.getElementById('firstName');
```

```
const lastName = document.getElementById('lastName');
```

```
const birthday = document.getElementById('birthday');
const username = document.getElementById('username');
const email = document.getElementById('email');
const phone = document.getElementById('phone');
const country = document.getElementById('country');
const terms = document.getElementById('terms');
```

```
const firstNameError = document.getElementById('firstNameError');
const lastNameError = document.getElementById('lastNameError');
const birthdayError = document.getElementById('birthdayError');
const usernameError = document.getElementById('usernameError');
const emailError = document.getElementById('emailError');
const phoneError = document.getElementById('phoneError');
const genderError = document.getElementById('genderError');
const countryError = document.getElementById('countryError');
const termsError = document.getElementById('termsError');
const successPrompt = document.getElementById('successPrompt');
```

```
const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
const phoneRegex = /^\d{10}$/;
```

```
firstName.addEventListener('focus', function() { firstNameError.style.display = 'none'; });
```

```
lastName.addEventListener('focus', function() { lastNameError.style.display = 'none'; });
```

```
birthday.addEventListener('focus', function() { birthdayError.style.display = 'none'; });
```

```
username.addEventListener('focus', function() { usernameError.style.display = 'none'; });
```

```
email.addEventListener('focus', function() { emailError.style.display = 'none'; });
```

```
phone.addEventListener('focus', function() { phoneError.style.display = 'none'; });
```

```
country.addEventListener('change', function() {  
    if (country.value !== "") {  
        countryError.style.display = 'none';  
    }  
});
```

```
terms.addEventListener('change', function() {  
    if (terms.checked) {  
        termsError.style.display = 'none';  
    }  
});
```

```
form.addEventListener('submit', function(event) {  
    event.preventDefault();  
    let isValid = true;  
    successPrompt.style.display = 'none';  
  
    if (firstName.value.trim() === "") {  
        firstNameError.style.display = 'block';  
        isValid = false;  
    } else {  
        firstNameError.style.display = 'none';  
    }  
}
```

```
if (lastName.value.trim() === "") {  
    lastNameError.style.display = 'block';  
    isValid = false;  
} else {  
    lastNameError.style.display = 'none';  
}
```

```
}
```

```
if (birthday.value === "") {  
    birthdayError.style.display = 'block';  
    isValid = false;  
} else {  
    birthdayError.style.display = 'none';  
}
```

```
if (username.value.trim().length < 4) {  
    usernameError.style.display = 'block';  
    isValid = false;  
} else {  
    usernameError.style.display = 'none';  
}
```

```
if (email.value.trim() === "" || !emailRegex.test(email.value.trim())) {  
    emailError.style.display = 'block';  
    isValid = false;  
} else {  
    emailError.style.display = 'none';  
}
```

```
if (!phoneRegex.test(phone.value.trim())) {  
    phoneError.style.display = 'block';  
    isValid = false;  
} else {  
    phoneError.style.display = 'none';  
}
```

```
const genderSelected =  
document.querySelector('input[name="gender"]:checked');  
  
if (!genderSelected) {  
    genderError.style.display = 'block';  
  
isValid = false;} else {genderError.style.display = 'none';}if (country.value === "")  
{countryError.style.display = 'block';isValid = false;} else  
{countryError.style.display = 'none';}if (!terms.checked) {termsError.style.display  
= 'block';isValid = false;} else {termsError.style.display = 'none';}if (isValid)  
{successPrompt.style.display = 'block';form.reset();});
```


Result/Conclusion

The program was executed successfully, and the required output was obtained.

